

Leaves

Concepts

- **Foliage leaves** are large, flat, green structures involved in photosynthesis.
- Natural selection has produced numerous types of leaves for protection (**bud scales**, **spines**), support (**tendrils**), storage (fleshy leaves), and nitrogen procurement (insect traps).
- Leaves are typically flat and thin for maximum absorption of light and CO_2 .
- Most leaf tissues are alive and differentiated into chlorophyll-rich chlorenchyma.
- Leaves typically live for only several months and contain only primary tissues; secondary production of wood and bark is rare.

External Structure of Foliage Leaves

- Functions of foliage leaves:
 - Photosynthesis: absorb CO_2 and convert it to carbohydrate using light energy.
 - Water conservation: must not lose excessive water.
 - Protection: against fungi, bacteria, epifoliar algae, and herbivores.
- **Leaf blade** (or **lamina**): the flat, light-harvesting portion.
 - **Dorsal surface** (abaxial): lower side, often with protruding veins.
 - **Ventral surface** (adaxial): upper side, usually smooth.

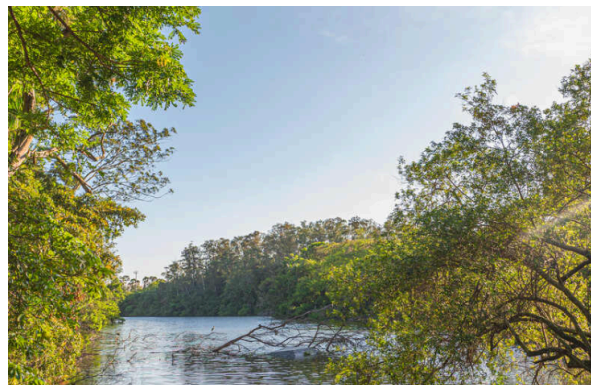


Figure 1: Several possible leaf shapes, all with equal volumes. (A) The common type with a thin, flat blade. All cells are exposed to light—there is a large surface area that absorbs CO_2 , and none of the veins are long. (B) In a thick leaf, light cannot penetrate to the bottom of the leaf, and those cells cannot photosynthesize. Relative to the volume, little surface area is available for CO_2 absorption, but this helps conserve water. (C) A spherical leaf has maximum internal self-shading, and only a small fraction of the cells receives enough light for photosynthesis. Surface area for gas exchange is minimal. (D) A cylindrical leaf could be thin with a large surface area and no self-shading, as in (A); however, to have the same volume as (A), it would be extremely long, and conduction through the veins might be difficult.

- **Petiole**: the stalk that holds the blade out into the light, preventing self-shading.
 - Allows blade flutter to cool the leaf and bring fresh air (replenishing CO_2).



Figure 2: The petioles of each leaf of poplar (*Populus alba*) are long enough that the leaves do not shade each other, and all are fully exposed to light. The lower (abaxial) side of each leaf blade is covered with white, dead trichomes, the upper (adaxial) surface is hairless and green.



Figure 3: Movement of Carbon Dioxide. As any molecule diffuses, it collides with other molecules and changes its direction of motion. Although air is almost entirely empty space, a molecule of carbon dioxide undergoes 9.4 billion collisions per second and moves only $0.039 \mu\text{m}$ each time.

- **Sessile leaf:** lacks a petiole; common in small or long narrow leaves where self-shading is not a problem (e.g., *Aeonium*).

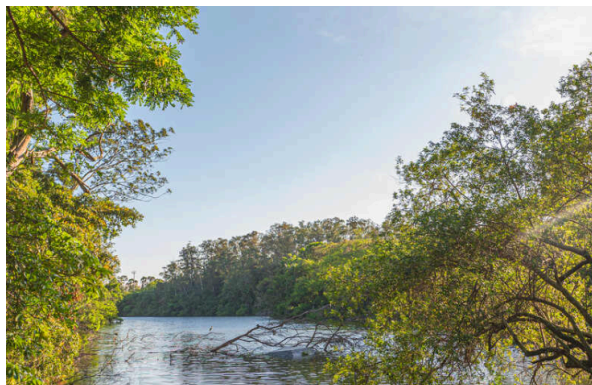


Figure 4: *Aeonium tabuliforme* and its relatives grow in regions of intense sunlight; self-shading is not a problem, but water conservation is. Close packing minimizes water loss from stomata.

- **Sheathing leaf base:** leaf base wraps around the stem; common in monocots (grasses, lilies, etc.).



Figure 5: (A) Long, narrow leaves are common in many monocots of intensely sunny areas. Shading is no problem, although leaf bases probably do not photosynthesize at maximum efficiency. Sotol (*Dasylirion*). (B) These are the leaves of St. Augustine grass (*Stenotaphrum*); the blades project horizontally and are attached to the stem by sheathing leaf bases. (C) In cattails (*Typha*), the sheathing leaf base can be half a meter long and provide firm but flexible attachment to the stem.

- Broad leaves in monocots: some have petiole-like structures (e.g., *Maranta*), but may be modified laminas.

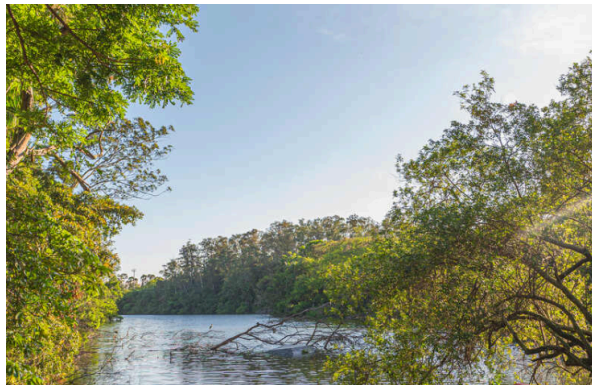


Figure 6: *Maranta*, a common houseplant, is a monocot that has broad leaves.

- **Simple leaf:** blade consists of just one part.

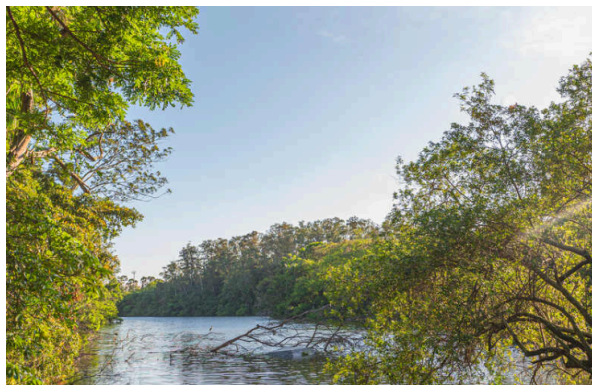


Figure 7: Simple leaves have numerous shapes, some so deeply lobed as to be almost compound. Poplar (A) and oak (C) have one dominant midrib from which lateral vascular bundles extend. In castor bean (B), petiolar bundles immediately diverge into several main veins.

- **Compound leaf:** blade divided into several individual parts called **leaflets**.
 - **Petiolule:** stalk attaching a leaflet to the **rachis** (extension of the petiole).

- **Palmately compound:** all leaflets attached at the same point.
- **Pinnately compound:** leaflets attached individually along the rachis.
- Advantages: prevents tearing by wind, increases turbulence (bringing in CO_2), and acts as a barrier to pathogens.



Figure 8: Leaflets of compound leaves have the same structure and metabolism as simple leaves. In pinnately compound leaves (rose, A) the petiolules of leaflets are attached to the rachis, but in palmately compound leaves (Virginia creeper, B) they attach to the end of the petiole. In either type, any leaflet can be damaged or can abscise without affecting the remainder of the leaf.

- **Doubly compound leaf:** leaflets themselves are compound (e.g., *Mimosa*).



Figure 9: This is a portion of one doubly compound leaf of *Mimosa*: The central rachis (midrib) bears numerous leaflets, but each leaflet itself is pinnately compound.

- **Heterophylly:** different leaf forms on the same plant (e.g., aquatic vs aerial leaves in *Myriophyllum*).



Figure 10: *Myriophyllum heterophyllum* grows immersed in flowing streams; its leaves are very delicate and pinnately compound with thread-like leaflets (leaflets have no lamina). If the stream is quiet, some stems emerge into the air and produce thicker, tougher leaves (not present here), hence the species name “heterophyllum.”

- **Economic importance and defense:**

- Leaves provide food (cabbage, lettuce) and flavorings (herbs, spices).
- Flavors often due to antiherbivore chemicals (e.g., alkaloids like nicotine in tobacco).

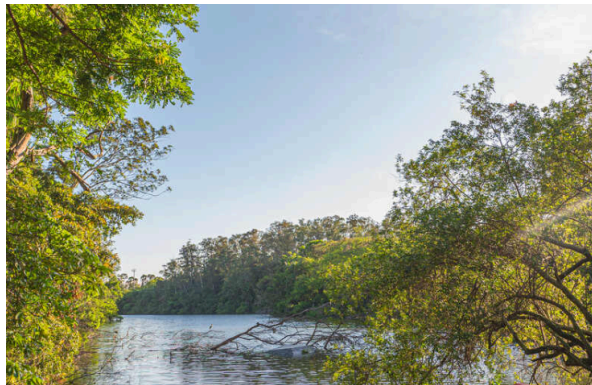


Figure 11: (A) Basil (the dots are glands that provide its flavor). (B) Spearmint. (C) Rosemary. (D) Tobacco.

- Leaf shapes vary widely and are used for identification.

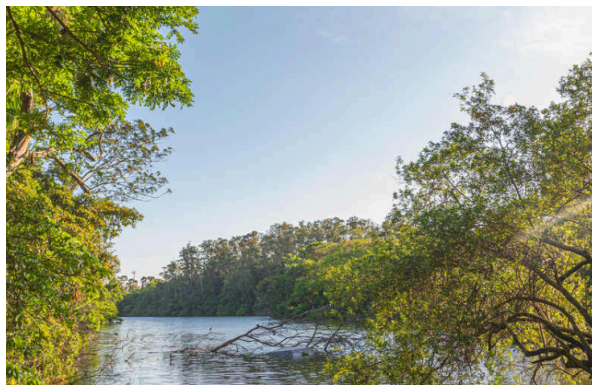


Figure 12: Four common leaf shapes; numerous other types also exist.

- **Heteroblasty:** juvenile leaves differ from adult leaves (e.g., beans, ivy).



Figure 13: (A) Beans have two types of leaves: The very first leaf formed by the seedling is simple (right), but all later leaves are compound (left). (B) This *Azara lanceolata* shoot appears to have two types of leaves: The larger leaves are true leaves, whereas the smaller ones are actually enlarged stipules.

- Leaf margins: can be entire, toothed, lobed, etc.

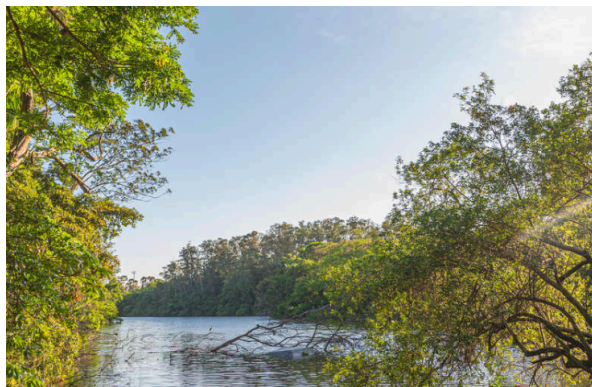


Figure 14: Several common types of leaf margins; in nature, hundreds of types exist, many being intermediate between two or among several other types.

- Venation patterns:
 - **Reticulate venation:** netted pattern, common in basal angiosperms and eudicots.

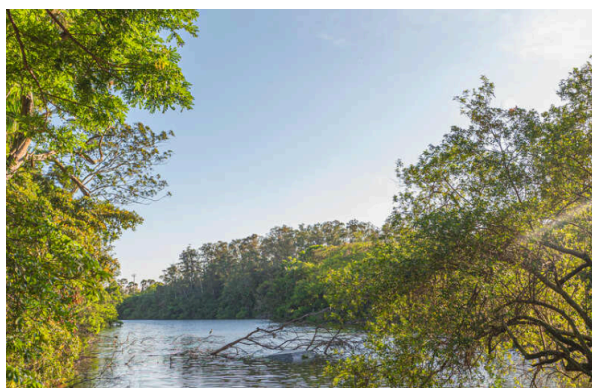


Figure 15: A fig leaf (*Ficus*) held up to the light reveals five prominent main veins, numerous lateral veins and the reticulate venation of all smaller veins in this eudicot leaf; no part of the leaf is without veins.

- **Leaf clearings:** reveal high density of veins; every cell is close to a vein.



Figure 16: These leaves were treated (cleared) to make their cells transparent; then xylem was stained to make leaf veins visible. (A) In passionflower (*Passiflora*) leaves, veins are narrow and highly branched. At this magnification ($\times 40$), the height of the photograph represents 1 mm, and veins occur close together. Leaf veins typically occur closer together than do capillaries in our bones, tendons, and cartilage. (B) Higher magnification of privet leaf. At any point, minor veins are only two or three tracheary elements wide ($\times 150$). (C) These minor veins of crown of thorns (*Euphorbia millii*, a species that grows in deserts) are broad and have many wide tracheids at the ends ($\times 50$). (D) *Acalypha* has numerous crystals within the leaf and trichomes on the leaf surface ($\times 80$).

- **Parallel venation:** larger veins run side by side, common in monocots.

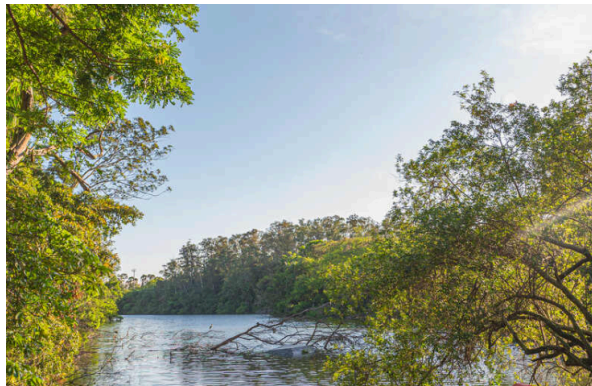


Figure 17: Cattails (*Typha latifolia*) have long, strap-like leaves common in monocots. Vascular bundles run parallel to each other from leaf base to tip.

- **Cross-veins:** fine transverse veins allow conduction if longitudinal veins are broken.



Figure 18: (A) Leaf clearing of corn (*Zea*), a monocot, with several sizes of longitudinal, parallel veins. If a longitudinal vein is broken by insect damage or any other problem, conduction can detour around the site by means of the fine transverse veins ($\times 50$). (B) Higher magnification showing the parenchyma cells that surround the leaf bundles in corn ($\times 250$).

- **Abscission zone:** oriented perpendicular to the petiole at the leaf base.
 - Cells release enzymes to weaken walls, allowing leaf to fall.
 - **Leaf scar:** protective corky tissue formed across the wound.

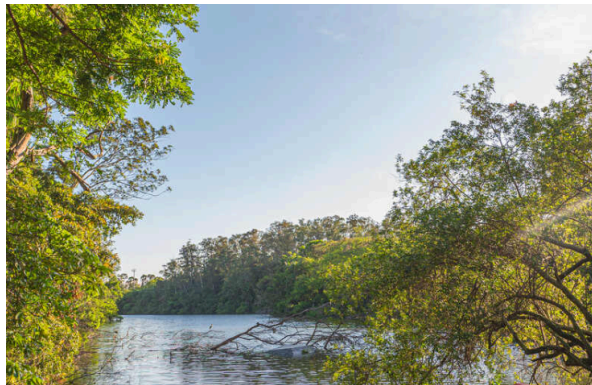


Figure 19: (A) Longitudinal section through a leaf axil, showing the abscission zone in the petiole and the axillary bud just above ($\times 15$). (B) Magnification of the abscission zone while the leaf is still healthy. The xylem and phloem are not interrupted until the leaf actually falls away ($\times 80$).

- Non-abscising leaves: dead leaves persist (e.g., Joshua tree), forming a protective skirt.



Figure 20: As leaves of Joshua tree (*Yucca brevifolia*) age, they flex downward and do not abscise; this creates a sheath of tough, fibrous dead leaves with many sharp spines that prevent rats, mice and other small animals from climbing the plant and eating young leaves and buds.

Internal Structure of Foliage Leaves

Epidermis

- **Transpiration:** water loss through the epidermis.
- Structure: flat, tabular cells, guard cells (stomata), and trichomes.
- Stomata distribution: usually more abundant on the lower epidermis to minimize water loss (avoiding convection currents on the heated upper surface).

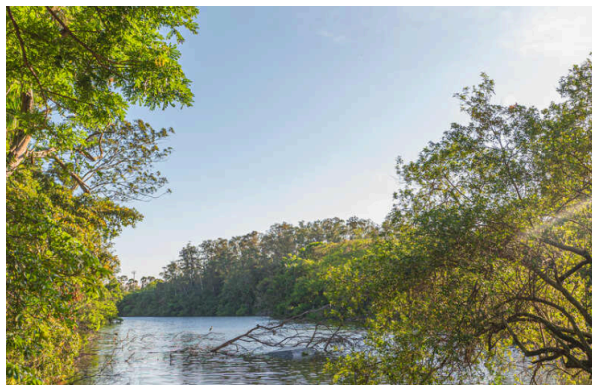


Figure 21: Frequency of Stomata in Leaf Upper and Lower Epidermis.

- Stomatal crypts: sunken cavities with trichomes (e.g., oleander) to trap moist air and reduce water loss.



Figure 22: (A) Oleander (*Nerium oleander*) leaves have stomatal crypts filled with trichomes and stomata. This minimizes air movement near stomata, and water molecules that diffuse out of the leaf may re-enter it by random motion. Crypts occur only in the lower surface ($\times 80$). (B) Each stoma of this grass (*Elytrostrachys*) is overarched by four protrusions from surrounding cells. This reduces air movement near the stoma ($\times 940$). (C) In this bamboo (*Chusqua*), stomata are protected by protrusions and wax rods ($\times 940$). In all three examples, the trichomes and waxes prevent water droplets from blocking the stomatal pores.

- **Trichomes:**

- Provide shade (deflect sunlight).
- Prevent rapid air movement (slow water loss).
- Deter insects (walking difficulty, stinging compounds).

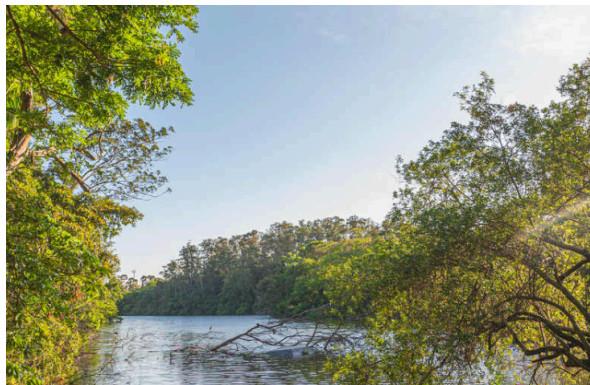


Figure 23: (A) Leaves of lamb's ears (*Stachys*) are so densely covered with hairs that they often appear white. The hairs reduce the amount of heat these leaves absorb from sunlight. (B) Plants of *Sempervivum* are native to high alpine regions where ultraviolet radiation is intense. The hairs of the leaf epidermis reflect part of the ultraviolet radiation away from the plant.

- Cuticle and wax: retain water and prevent fungal digestion.

Mesophyll

- **Mesophyll:** ground tissues interior to the epidermis.
- **Palisade parenchyma** (palisade mesophyll):
 - Usually along the upper surface.
 - Main photosynthetic tissue.
 - Cells separated slightly for maximum surface area for CO_2 absorption.



Figure 24: This leaf of *Laurelia* has a layer of palisade parenchyma that is two cells deep, and the spongy mesophyll is an extensive aerenchyma ($\times 180$).

- Surface area: slight separation of cells increases area for CO_2 absorption.

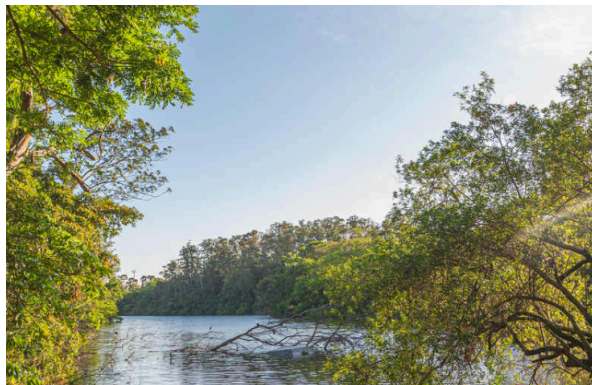


Figure 25: (A) If palisade cells are closely packed with no space between them, only their bottoms are exposed to the carbon dioxide in the spongy mesophyll. (B) With only slight separation of the cells, the volume of the palisade is almost unchanged, but the surface area available is increased enormously; carbon dioxide can be absorbed much more rapidly. Assume that the palisade parenchyma cells are rectangular columns $20 \times 20 \times 100 \mu\text{m}$ long; compute the surface area exposed to CO_2 by 100 cells arranged as in (A) versus (B).

- Thickness: can be multiple layers in high light (e.g., holly).

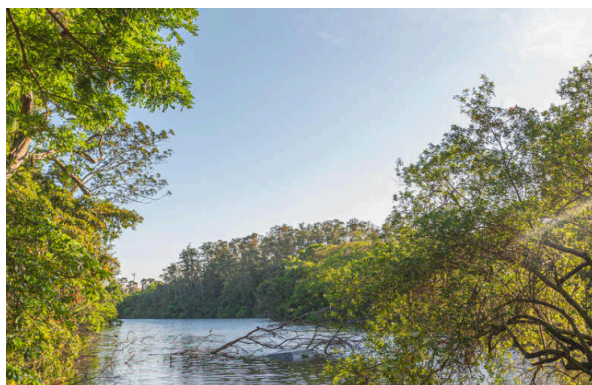


Figure 26: (A) and (B) Leaves of holly (*Ilex*) have an extremely thick palisade parenchyma consisting of at least three layers of columnar cells. Cells of the lowest palisade layer are relatively widely separated, permitting rapid diffusion of CO_2 from spongy mesophyll to upper palisade (A) $\times 50$; (B) $\times 150$).

- **Spongy mesophyll:**

- Located in the lower portion.
- Open, loose aerenchyma permitting rapid CO₂ diffusion.
- Some plants have palisade parenchyma on both sides (e.g., vertical leaves).

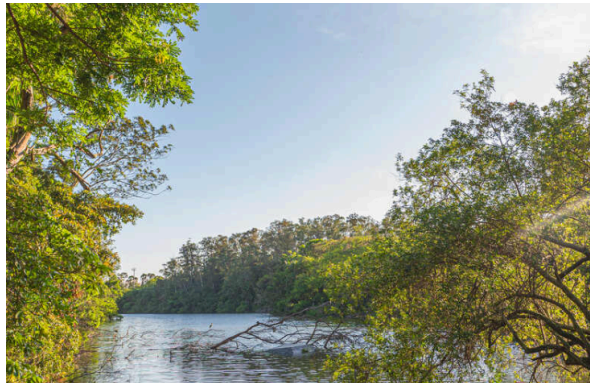


Figure 27: Leaves of both carnation (A) and creosote bush (B) have palisade parenchyma along both surfaces. The little spongy mesophyll present is located in the middle of the leaf (A) $\times 75$; (B) $\times 150$).

Vascular Tissues

- **Midrib** (midvein) and **lateral veins**: contain primary xylem (upper) and primary phloem (lower).
- **Bundle sheath**: fiber sheath around vascular tissues for support and protection.

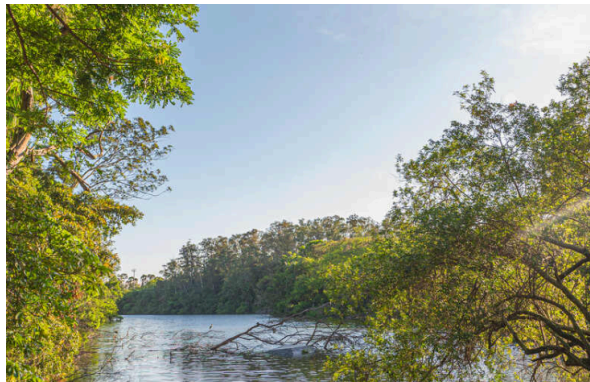


Figure 28: Transverse sections through leaf midribs. (A) This midrib of *Ligustrum* is rather simple, containing a single vascular bundle. Xylem is present along the top, phloem along the bottom. A small amount of collenchyma occurs on the top and bottom of the midrib ($\times 150$). (B) This midrib of rubber tree has numerous separate bundles embedded in a large mass of mesophyll ($\times 50$). (C) This is a midrib prepared as a leaf clearing (similar to Figure 6-14); the three distinct bundles have no interconnections in this region. Each transports to and from a particular portion of the blade ($\times 15$).

- **Bundle sheath extension**: fibers extending to epidermis, providing support and conducting water to mesophyll.



Figure 29: (A) This leaf bundle of sugar cane has bundle sheath extensions contacting both the upper and lower epidermis. Water diffuses from the xylem to the epidermis by capillary action, moving around the surfaces of the fibers ($\times 100$). (B) The leaves of flax have large bundle sheath extensions consisting of many layers of fibers ($\times 45$).

- **Minor veins:** sites of material exchange; large surface area in contact with mesophyll; no fibers.

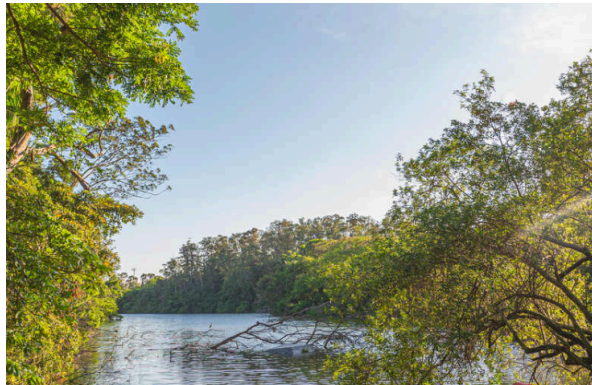


Figure 30: (A) and (B) These are minor veins, treated to reveal individual cells. Long, dark cells are tracheary elements, and phloem is out of focus, in line with the xylem. Notice how narrow these veins are: Xylem is only one element wide, so there is maximum surface area for releasing water to the leaf mesophyll. In this species, each minor vein is lined with parenchyma cells that extend outward into the spongy mesophyll ($\times 250$).

- **Leaf Structure Review** (Box 6-2):
 - Palisade parenchyma: cylindrical, not closely packed.
 - Veins: network below palisade.
 - Spongy mesophyll: extensive air space.
 - Lower epidermis: abundant stomata.



Figure 31: (A) Palisade parenchyma ($\times 250$). The cells are cylindrical and are not closely packed; carbon dioxide diffuses easily throughout this layer and is absorbed into the cells through their large surface area. (B) Leaf veins ($\times 50$). Just below the palisade parenchyma lies the network of leaf veins. Small lateral veins branch off of larger veins and in turn give off smaller veins. (C) High magnification of a large lateral vein ($\times 250$). This section passes through the xylem of the vein, and perforations are visible; thus, these are vessel elements, not tracheids. These have spiral thickenings: They could differentiate while the leaf was still small, and then as the leaf expanded, they could be stretched and still conduct water. (D) High magnification of minor veins ($\times 250$). All mesophyll cells are close to one vein or another. After sugar has been produced by photosynthesis, it does not diffuse very far before it is picked up by a sieve tube member and transported to the stem. (E) Spongy mesophyll ($\times 250$). In many leaves, more than half the spongy mesophyll consists of intercellular air space. This allows carbon dioxide molecules to diffuse rapidly into the leaf, away from the stomata. If spongy mesophyll were more dense, carbon dioxide molecules might be trapped near the stomata, then diffuse back out of the leaf before they could be absorbed by a chlorenchyma cell. (F) Lower epidermis ($\times 150$). The epidermis is slightly undulate, and as it was cut to make this slide, the knife passed through the center of the cells in some areas (green) and through the outer wall and cuticle in other areas (red). Stomata are abundant, but the actual stomatal pore is short and narrow; each is only a small opening through which carbon dioxide can enter.

Petiole

- Transition between stem and lamina.
- **Petiole vascularization:** vascular bundles (leaf traces) branch from stem and enter petiole; patterns vary (ring, plate, etc.).



Figure 32: (A) This section of stem has 11 bundles, 3 of which produce leaf traces that enter the petiole. Later, the leaf traces diverge and spread throughout the blade. These traces are drawn without any branching or merging, although both are possible. (B) Cross-section of a petiole of the palm *Attalea* stained to show its many vascular bundles. Notice how numerous the bundles are compared with the leaves of Figure 6-25. This leaf, like those of most palms, is very large and must have a massive conduction system. (C) In cacti, traces that run to leaves and axillary buds are complex partly because the cortex is so thick the traces must be long, and partly because leaf and branch traces divide and bifurcate, forming an extensive set of cortical bundles that vascularize the thick, succulent cortex. The central stele of the cactus is the dark material on the right; the dark, thick traces at the top of the micrograph are leaf and bud traces that lead to a flower, and all the others are cortical bundles ($\times 3$). *Hylocereus undatus*.

- **Stipules:** small flaps of tissue at the petiole base; may protect the shoot apical meristem.

Initiation and Development of Leaves

Basal Angiosperms and Eudicots

- **Leaf primordium:** protrusion from the shoot apical meristem.

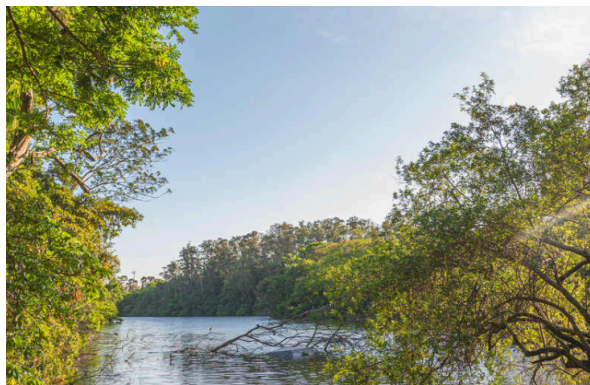


Figure 33: The shoot apical meristem of grape (*Vitis riparia*) can be seen in three dimensions in this scanning electron micrograph. Notice the large leaf primordium; an older leaf primordium was removed; the phyllotaxy is established at the very apex ($\times 200$).

- **Development:**
 - Primordium grows upward as a narrow cone.
 - Midrib establishes thickness.
 - Marginal meristems initiate the lamina (wings).
 - In compound leaves, leaflet primordia are initiated.



Figure 34: (A) The uppermost leaf primordium is the oldest, and its sides have begun growing out as the lamina of a simple leaf of grape ($\times 270$). (B) In this compound leaf, rather than forming a lamina, six leaflet primordia have been established ($\times 150$).

- Cellular activity: cells are small, cytoplasmic, dividing; veins differentiate.



Figure 35: (A) Low-magnification transverse section through young, developing leaves of the grass *Panicum effusum*. The central round structure is a section of the shoot apical meristem, and the leaves encircle it (grasses have sheathing leaf bases) ($\times 200$). (B) At higher magnification, you can see that all cells in the leaves are still small and cytoplasmic, and the central vacuoles are small. At this stage, all cells are still undergoing mitosis and cytokinesis. Veins are starting to differentiate ($\times 500$). (C) At an older stage, leaf veins are more distinct. The first protoxylem tracheary elements are mature in some veins, but most cells that will differentiate into tracheary elements are still enlarging ($\times 500$).

- Expansion:
 - In perennials, leaves may be initiated in autumn and expand in spring.
 - Expansion involves water absorption and cell swelling.



Figure 36: Buds of buckeye (*Aesculus arguta*) expanding in the spring. Bud scales are being pushed back and will abscise as new foliage leaves swell. The smaller bud on the right contains only leaves, and its apical meristem may be initiating new leaf primordia that will expand a little later. The larger bud on the left has both new leaves as well as flower primordia (the small ball-like structures).

- Vulnerability: young leaves lack sclerenchyma and are vulnerable to insects (e.g., aphids).

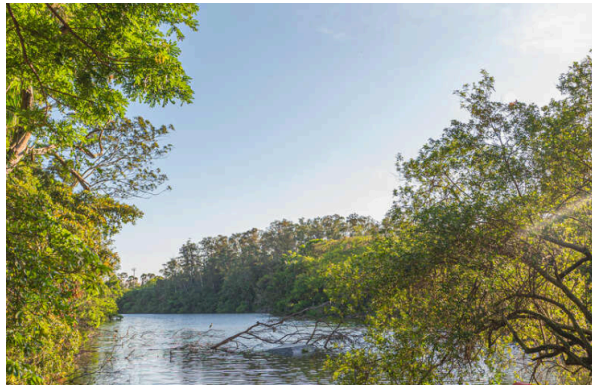


Figure 37: Aphids and other sucking insects typically attack young leaves and stems that have not yet developed any protective sclerenchyma around the vascular bundles.

Monocots

- Leaf primordium encircles the shoot apical meristem to form a sheathing leaf base.
- Lamina grows from a basal meristem (intercalary meristem) at the top of the sheath.
- Allows continuous growth (e.g., grasses) even if the tip is removed.



Figure 38: In a monocot, after the center of the leaf primordium has been initiated (A), neighboring regions of the shoot apical meristem become active and contribute to the primordium. This results in (B) a leaf base that encircles the stem, forming a sheath around it (A) $\times 150$; (B) $\times 120$). (C) The leaf blade of this *Dendrobium* orchid has developed from a leaf primordium tip that resembled the tip in (B), and the sheathing base here developed from elongation growth of the small collar of leaf primordium tissue visible in (B).

• **Photosynthesis Without Leaves** (Box 6-3):

- **Photosynthetic bark:** green bark with chlorophyll (e.g., palo verde); “recycling” photosynthesis.
- **Photosynthetic cortex:** common in young stems and stem succulents (cacti).
- **Stem succulents:** store water, have stomata in stem epidermis, and cortical bundles.

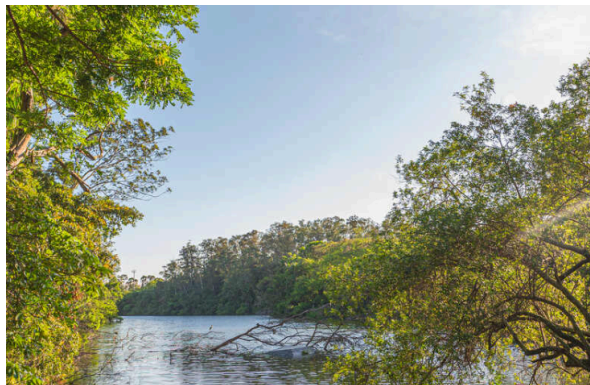


Figure 39: (A) Trunks and branches of trees called “palo verde” (green stick; *Cercidium floridum*) are green because their bark contains cells with chlorophyll. No stomata are present, so no new carbon dioxide can be absorbed from the air. (B) Trunks and branches of saguaro cactus (pronounced sa-WA-row there is no “g” sound at all) are green because the epidermis is still present and transparent, allowing us to see the green of the cortex cells inside. Stomata are present, and saguaros (*Carnegiea gigantea*) do bring in carbon dioxide from the air and convert it to carbohydrate. As a saguaro absorbs water after a rain, its parenchyma cells swell so the entire stem swells; the ribs flatten out and the epidermis is not torn. If it does not rain for many months, the cactus gradually loses water, despite its cuticle and wax, and shrinks; the ribs also shrink and become narrower (think of an accordion letting out air). A ribbed body like this can change volume without changing surface area.

Morphology and Anatomy of Other Leaf Types

Succulent Leaves

- Adapted for water storage in desert habitats.
- Thick and fleshy with reduced surface-to-volume ratio.
- Mesophyll has few air spaces (reduces water loss, increases transparency).
- Examples: Crassulaceae, *Lithops*.



Figure 40: (A) Leaves of *Senecio rotundifolia* are spherical, giving them an optimal surface-to-volume ratio for conserving water. (B) Leaves of the succulent *Dinteranthus* are hemispherical and attached to the opposite leaf, greatly reducing the exposed leaf surface. Interior tissues are mostly water-storage parenchyma. Their low levels of chlorophyll result in low levels of photosynthesis, but a lack of water is a more significant danger for these plants. Each plant consists basically of two leaves and a microscopic stem; the root system (not visible) is substantial. *Dinteranthus* and many related species have only two leaves at a time; as two new leaves expand, the existing two collapse and wither away. (C) This agave (probably *Agave parryi*) has survived a wildfire in West Texas because its succulent leaves protected the stem from the heat of the fire. The center of the plant—and especially the shoot apical meristem—is still healthy. Containing so much water, the leaves heat up slowly, and before the innermost leaves became hot, wind had blown the fire farther on and the stem and apical meristem were saved. The leaves even kept the ground at the base of the plant cool, protecting the roots.

- **Window leaves:** translucent tips conduct light to subterranean chlorenchyma (e.g., *Lithops*).



Figure 41: *Lithops*, stone plant, is closely related to *Dinteranthus*; both are in the family Aizoaceae. *Lithops* also has only two leaves, but they are located almost entirely underground. The flat, translucent tips project above the soil and conduct light to the subterranean chlorenchyma.

Sclerophyllous Foliage Leaves

- **Sclerophylls:** tough, hard leaves with abundant sclerenchyma (fibers).
- Perennial leaves (2+ years).
- Resistant to animals, fungi, freezing, and UV light.
- Examples: barberry, holly, *Yucca*.

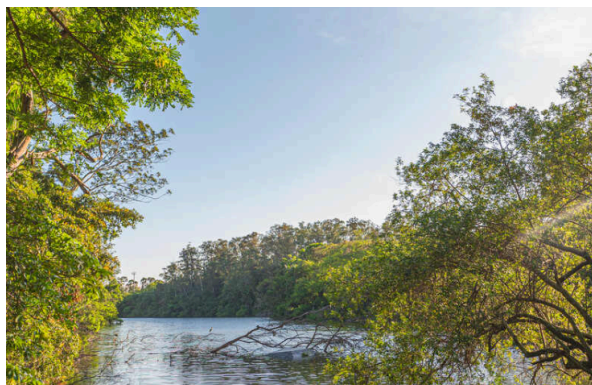


Figure 42: Leaves of barberry (*Berberis*) are tough and hard. It is difficult for insects to bite into them or lay eggs in them; however, the plant must invest considerable glucose to make the secondary walls of sclerenchyma cells, and the leaves must therefore photosynthesize longer before they reach their break-even point.

- Structure: thick cuticle, thick-walled epidermis, fiber bundles, fiber sheaths.



Figure 43: (A) *Yucca* leaves are excellent examples of sclerophylls; they are tough and fibrous, and the cuticle is thick. These features reduce the amount of light reaching the chlorenchyma, but plant growth is probably limited by scarcity of water rather than reduced photosynthesis. These leaves may function for 5 to 10 years or more ($\times 15$). (B) Bundles of fibers occur just interior to the epidermis, and the thick mesophyll surrounds many vascular bundles ($\times 50$). (C) Stomata occur in long, shallow grooves that run between fiber bundles ($\times 150$). (D) Even vascular bundles of these sclerophylls have fiber sheaths ($\times 150$). (E) Sclerophylls of *Dracaena* are very thick and have leaf veins throughout the mesophyll, unlike the single row in ordinary leaves ($\times 15$). (F) Leaf veins of *Dracaena* have extremely thick fiber sheaths ($\times 150$).

Leaves of Conifers

- Mostly sclerophylls with thick cuticle and thick-walled epidermis/hypodermis.
- Simple leaves: needles (pine, fir) or scales (juniper, cypress).
- Perennial (evergreen).

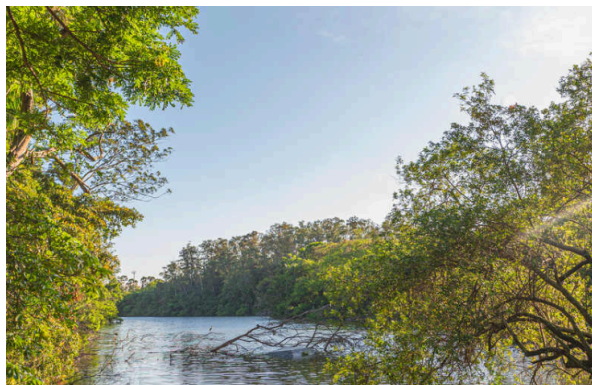


Figure 44: (A) Many conifers, such as this lodgepole pine (*Pinus contorta*), have needle-shaped leaves. (B) Incense cedar (*Libocedrus*) and many other conifers have scale-shaped leaves.

- Scale leaves: small, flat, shield-like (e.g., *Araucaria*).



Figure 45: Scale-shaped leaves in monkey-puzzle tree (*Araucaria araucana*) are very large and persistent, living for many years, even as the branch enlarges and becomes woody.

- Needle structure: round in section, resin canals, sunken stomata.

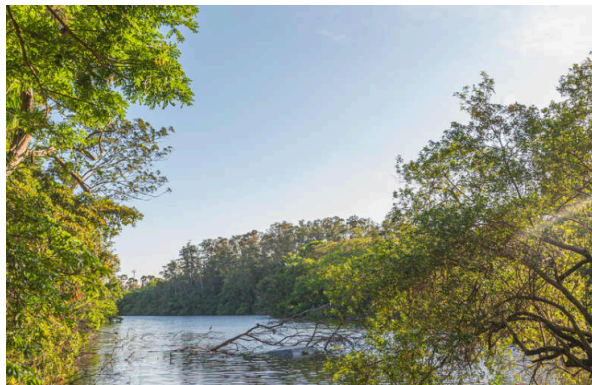


Figure 46: (A) Needle leaves of pine are round in transverse section ($\times 15$). (B) Needles last for many years, producing more phloem each year ($\times 150$). (C) Pine needles are sclerophylls: They have thick-walled epidermal cells with a thick cuticle. They also contain resin canals ($\times 150$).

Bud Scales

- Modified leaves protecting dormant shoot apical meristems.
- Small, tough, waxy, often corky; short or absent petiole.

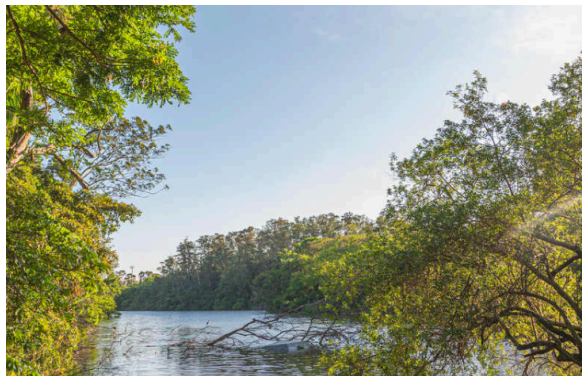


Figure 47: (A) Bud scales of *Rhododendron* fit together tightly during winter, protecting the enclosed apical meristem and young leaves and flowers. (B) The bud scales of *Magnolia* have a dense covering of hair; this improves the insulation.

Spines

- Modified leaves of axillary buds (e.g., cacti).
- Protective function; no blade; mesophyll consists of fibers.
- Photosynthesis carried out by stem cortex.



Figure 48: (A) The spines of this barrel cactus (*Ferocactus*) are modified leaves. The cluster of spines is actually an entire axillary bud. (B) Cactus spines have a basal meristem; as new cells are formed, older cells are pushed upward. As they move out of the meristem, they fill their central vacuole with water, elongate, then deposit a thick secondary wall and differentiate into fiber cells ($\times 50$).

Tendrils

- Modified leaves for support (e.g., peas).
- Grow indefinitely; sense contact and coil around objects.
- No lamina.

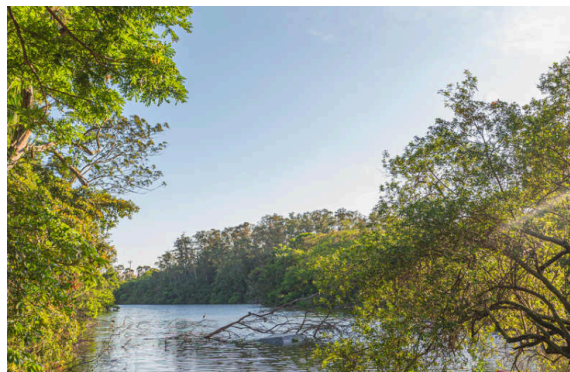


Figure 49: This tendril is a highly modified leaf: It has no leaf blade, and whereas foliage leaves stop growing after they reach a specific size, this tendril continues to grow.

Leaves with Kranz Anatomy

- Occur in C4 plants.
- Lack distinct palisade/spongy mesophyll.
- Prominent bundle sheaths with large chlorophyllous cells surrounded by a ring of mesophyll cells.

Insect Traps

- Adapted for nitrogen-poor habitats.
- Active traps (move): Venus' flytrap, sundew.
- Passive traps (no movement): pitcher plants.

- Digest insects to obtain nitrogen.

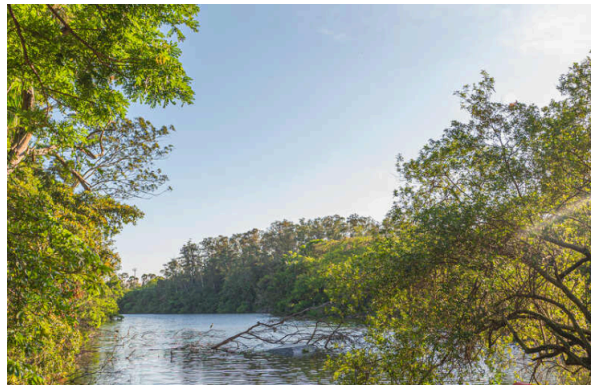


Figure 50: (A) *Nepenthes* has elaborate pitcher leaves. They have an ordinary petiole and lamina, but the leaf tip is long, narrow, and pendant. Its extreme tip turns upright and develops into the hollow pitcher that contains a digestive mixture and an epidermis that can absorb nitrogen. The ultimate portion of the leaf tip is a broad flat roof that prevents rain from falling into the pitcher and diluting the digestive juices. (B) Sundew leaves have many shapes; those of *Drosera capensis* are long and narrow. Just after an insect is caught, the leaf curls and places numerous stalked glands on it. The drop at the end of each gland is both sticky and digestive. Once digestion is complete, the leaf uncurls and is ready for its next meal. (C) Each half of the blade of a Venus' flytrap leaf has three hairs; if an insect touches any two, it stimulates the trap to close.